

Sample Inspection Report

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Inspection Report: Vxxxxx-2017-08-30

Wind field	Windy Heights
Wind turbine	V 90; Vxxxxx; V90, 2.0 MW, N 105
Blade-Serial #	S1, S2, S3

Field work date	30.Aug.2017
Inspection date	14.Mar.2018

Summary

Some laminate and coating defects have been found based on the exterior pictures of the rotor blades.

Two areas with lightning damages were detected on rotor blade B, pressure side, at radius Z17.3 and Z43. The laminate layers of the blade shell are obviously damaged. A profound repair is recommended.

On the outer leading edges of all three rotor blades coating defects caused by erosion and wear were detected. Most defects are located between radius Z37.5 and the tip (Z44).

Furthermore, a lot of vortex generators are damaged or completely missing. All these streaming elements should be replaced.

Repair within next 3 months is recommended.

Notice: Due to the camera perspective some areas (blade collar, end face of the trailing edge) could not be inspected exactly. Further defects in these areas cannot be excluded.

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Reviewed:	XXX (cp.max Rotortechnik)
Verified:	

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1 Glossary

1.1 Terms

Term or abbreviation	Definition
Blade root	A specific point on the longitudinal axis of the blade measured from the exact center of the hub. Note that the size of the hub is not the same in the various wind turbine types.
Leading edge (LE)	The leading edge of the blade, i.e. the 'nose' that is heading into the wind during operation.
Suction Side (SS)	In the manufacturing the SS shell is the upper shell. It is the side of the blade which faces the tower during turbine operation.
Shell	The function of the shell is to give an aerodynamic profile.
Spar	This is the structural part of the blade. It is positioned between the shells and must resist loads and forces. The sides towards the leading and trailing edge are called the web.
Trailing edge (TE)	The trailing edge of the blade is the thin edge where the airflow leaves the blade during operation. Together with the main spar the trailing edge is sustaining loads to the blade.
Pressure Side (PS)	The PS shell is the underside during production. This side is facing the wind and is also known as the pressure side.

1.2 Classification of Defects

Class	Description
Category 1	Cosmetic Damage No need for immediate action.
Category 2	Damage below wear and tear Repair only if other damages are to be repaired.
Category 3	Damage above wear and tear Repair must be done within next 6 to 12 months.
Category 4	Serious Damage Repair must be done within next 3 to 6 months.
Category 5	Critical Damage Immediate action required to prevent turbine damage. Contact technical support. Stop operation, safety is not ensured.

2 Introduction

2.1 Location and WTG

On 30. Aug. 2017 a blade inspection with AUS has been performed on WTG unit V 90, Vxxxxx in Windy Heights.

2.2 Technology

The inspection has been carried out with an autonomous UAS (Sulzer & Schmid Laboratories Technology), making high-resolution images of all blades.

2.3 Target

The aim of this report is to inspect in detail the surface of the blades and thus documenting its status. Where damages are detected, those are documented below. Recommendations are given if any actions should be taken.

2.4 Data concerning the case

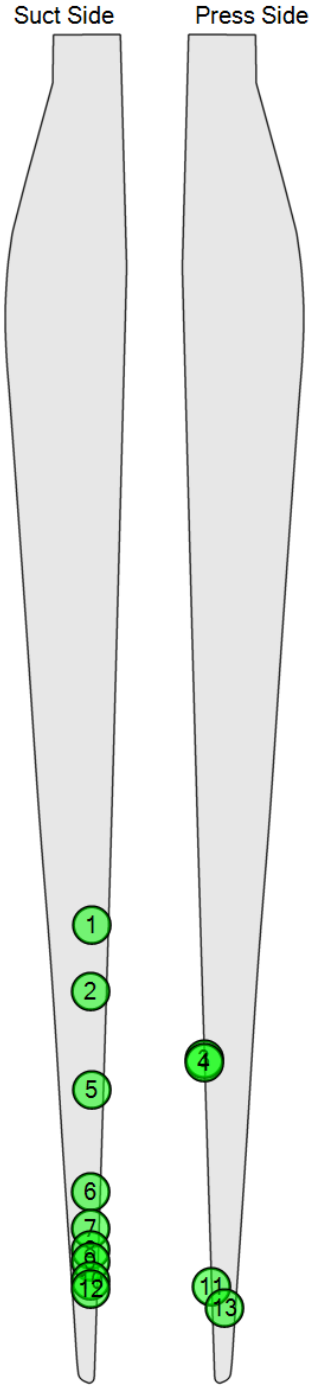
Team	
UAV Operating	D. Brändle
WTG Operator	XXX
Blade Expert	XXX (cp.max Rotortechnik)
Report Reviewer	XXX (cp.max Rotortechnik)
Report Verifier	
Unit	
Wind park	Windy Heights
Location Details	Zip City
Owner	Windy Heights Owner
WTG Type	V90, 2.0 MW, N 105
WTG No.	Vxxxxx
Blade Serial No.	S1, S2, S3
Commissioning Date	23.Mrz.2005
Operational Hours	0
Production kWh	0
Inspection Gear	
Drone	DJI, Matrice M600
Software	SSL3DX™

Weather	
Weather	sunny sky
Wind	2-4 m/s
Temperature	24°C

3 Blade Inspection

3.1 Blade A No. S1

3.1.1 Overview Blade A



Defect Number	Class	Type
1	Category 2	VG damaged
2	Category 2	VG missing
3	Category 2	Chip in paint/coat
4	Category 2	Chip in paint/coat
5	Category 2	VG damaged
6	Category 2	VG damaged
7	Category 2	VG damaged
8	Category 2	VG damaged
9	Category 2	VG damaged
10	Category 2	VG damaged
11	Category 2	Erosion
12	Category 2	VG missing
13	Category 2	Lightning discoloration

3.1.2 Details Blade A

Defect: 1

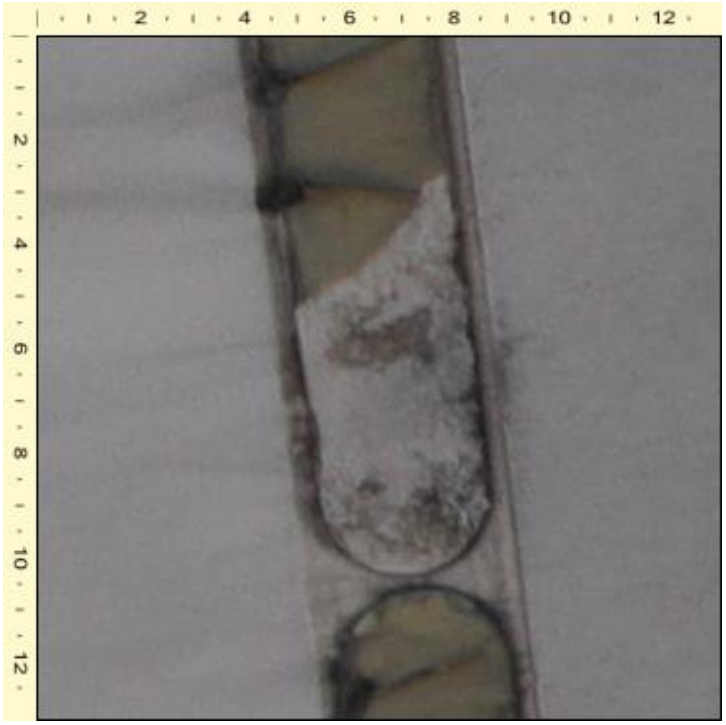


Image: 1
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 29m
Size: 8.3cm
Type: VG damaged
Class: Category 2
Description: L-size

Defect: 2

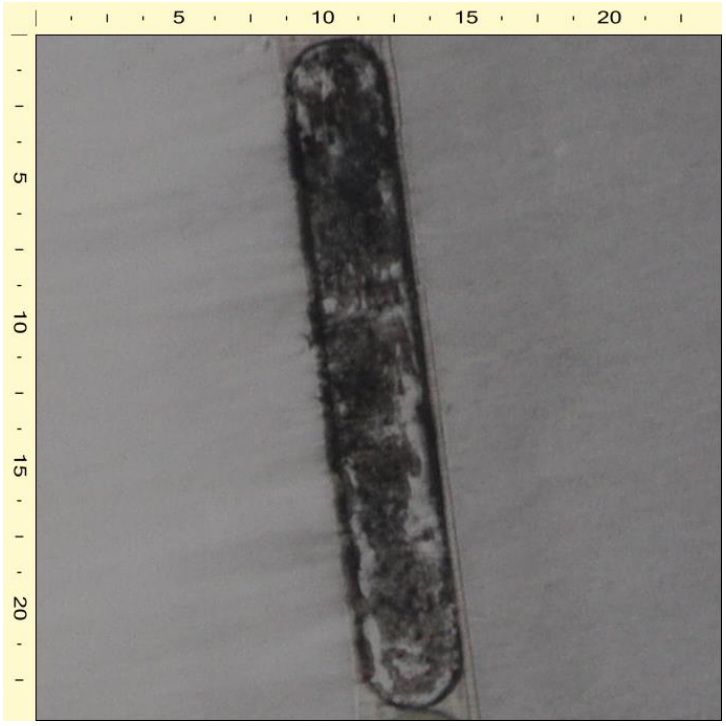


Image: 2
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 31.2m
Size: 22.9cm
Type: VG missing
Class: Category 2
Description: L-size

Defect: 3

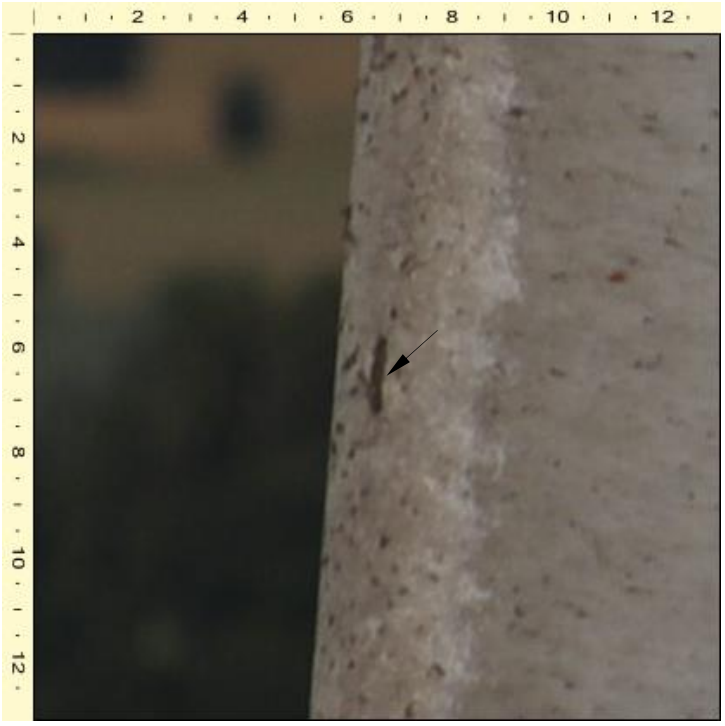


Image: 3
Blade: A
Blade Side: PS
Position: LE
Distance from root: 33.4m
Size: 1.6cm
Type: Chip in paint/coat
Class: Category 2
Description:

Defect: 4



Image: 4
Blade: A
Blade Side: PS
Position: LE
Distance from root: 33.5m
Size: 0.7cm
Type: Chip in paint/coat
Class: Category 2
Description:

Defect: 5

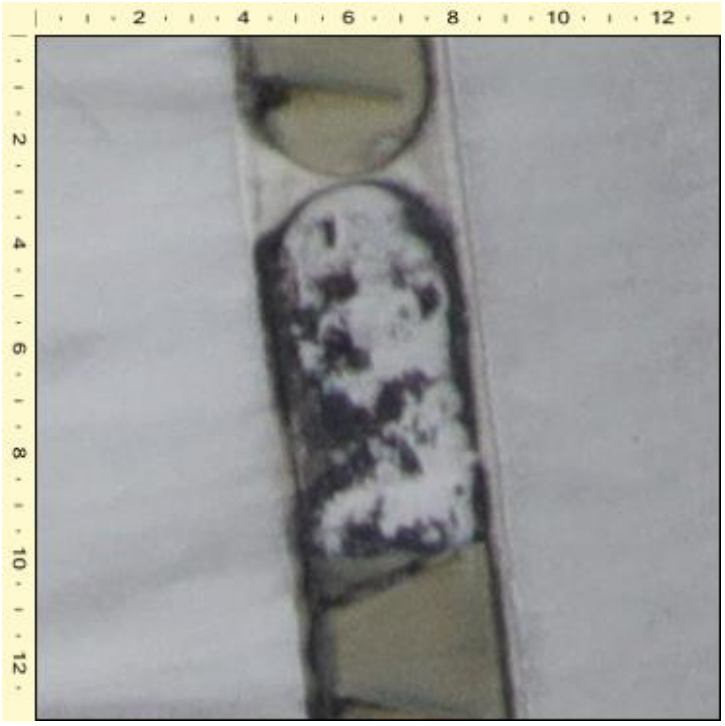


Image: 5
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 34.4m
Size: 7.7cm
Type: VG damaged
Class: Category 2
Description: L-size

Defect: 6

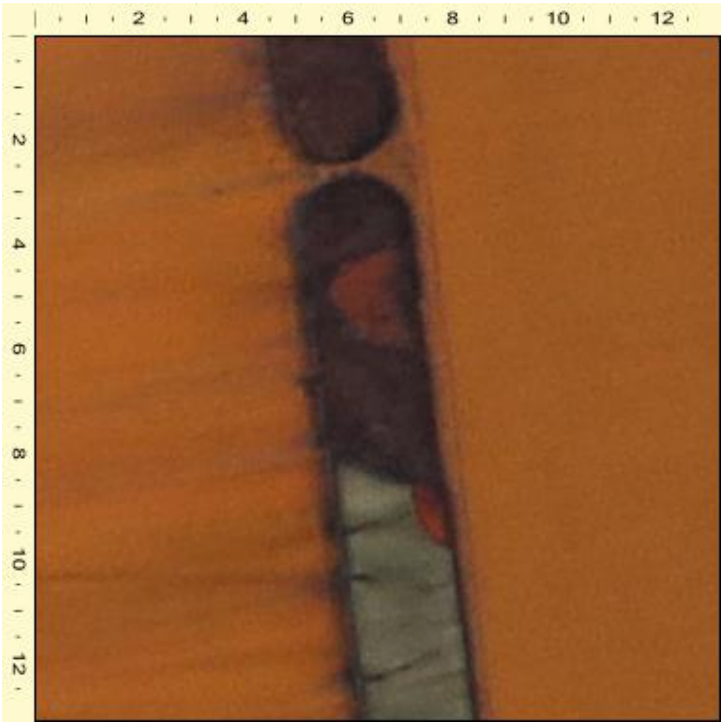


Image: 6
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 37.7m
Size: 74.1cm
Type: VG damaged
Class: Category 2
Description: S-size,
3 elements, damaged or missing

Defect: 6



Image: 7

Blade: A

Blade Side: SS

Distance from LE: 30%

Distance from root: 37.7m

Size: 74.1cm

Type: VG damaged

Class: Category 2

Description: S-size,
3 elements, damaged or missing

Defect: 7

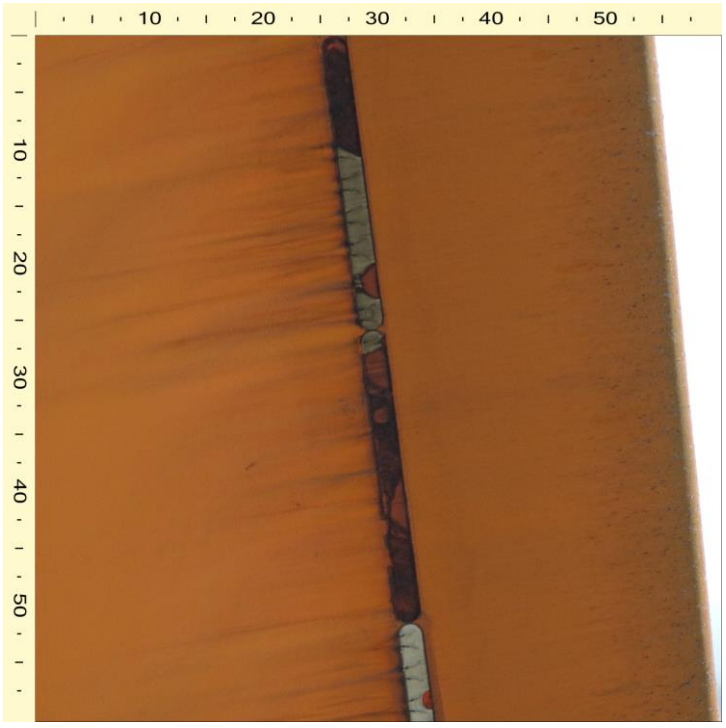


Image: 8

Blade: A

Blade Side: SS

Distance from LE: 30%

Distance from root: 38.9m

Size: 58.1cm

Type: VG damaged

Class: Category 2

Description: S-size, 3 elements

Defect: 8

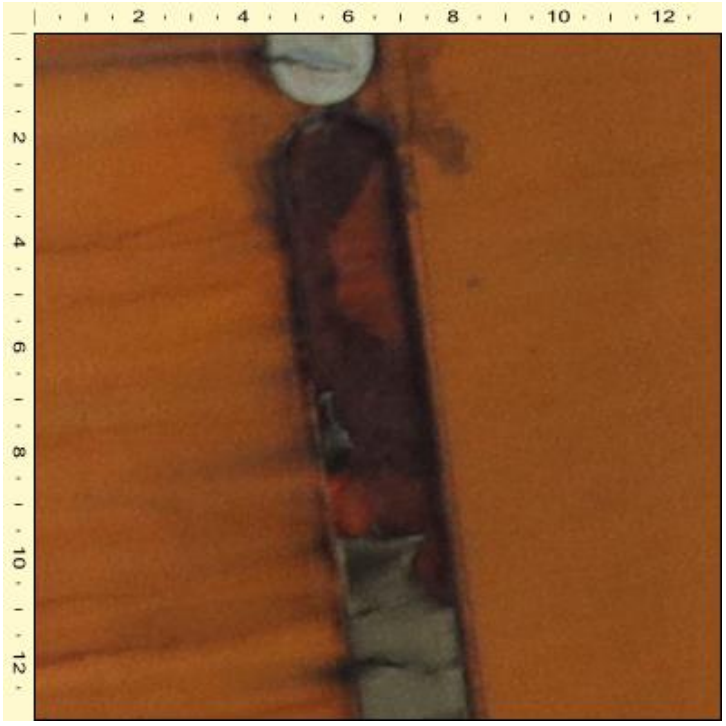


Image: 9
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 39.6m
Size: 10.1cm
Type: VG damaged
Class: Category 2
Description: S-size

Defect: 9

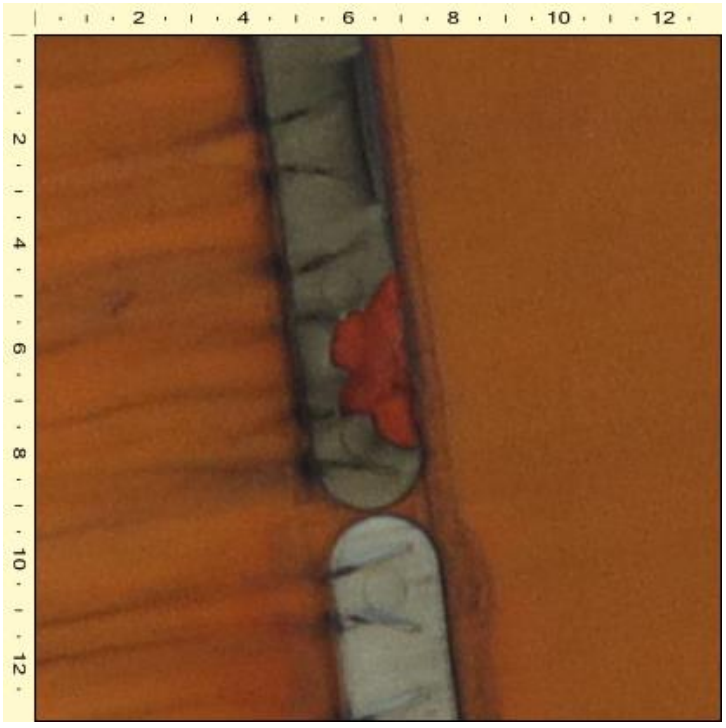


Image: 10
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 40m
Size: 4.9cm
Type: VG damaged
Class: Category 2
Description:

Defect: 10

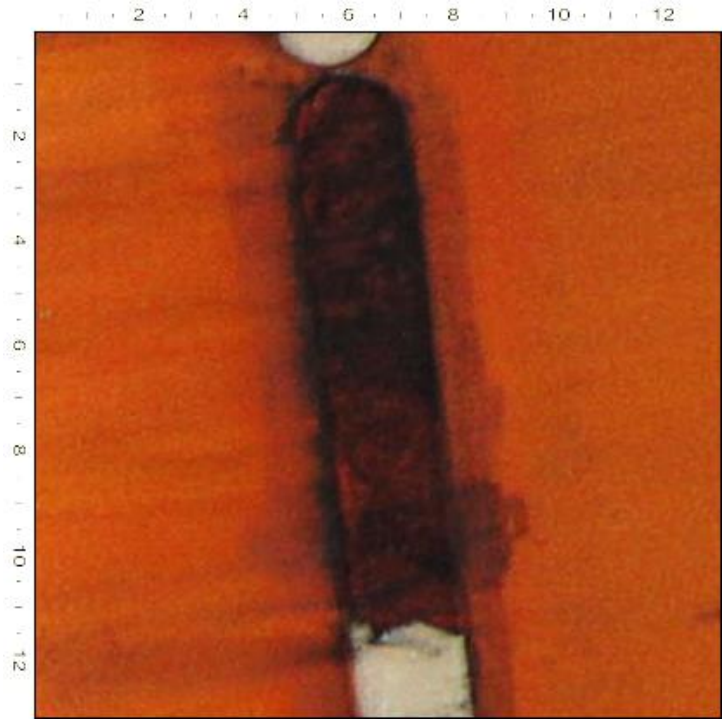


Image: 11
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 40.6m
Size: 10.8cm
Type: VG damaged
Class: Category 2
Description: S-size

Defect: 11

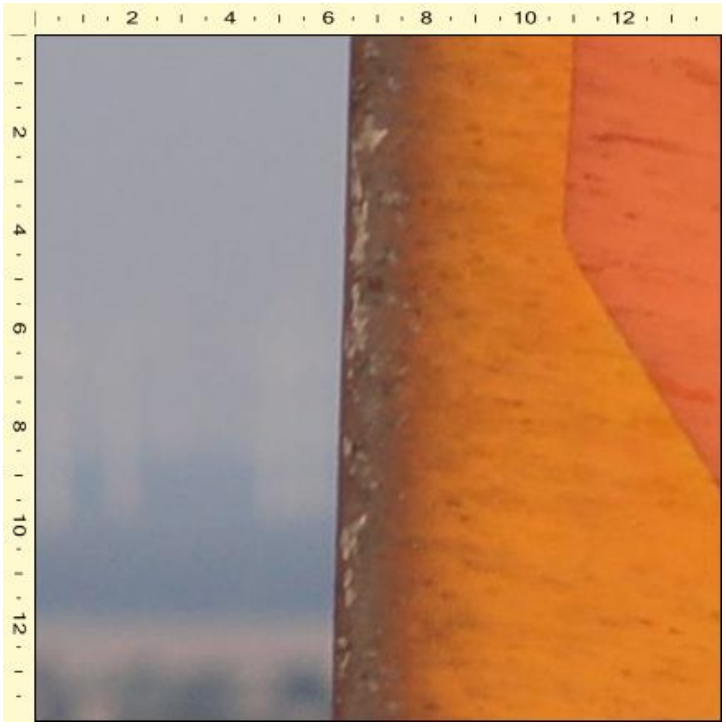


Image: 12
Blade: A
Blade Side: PS
Position: LE
Distance from root: 40.8m
Size: 13.2cm
Type: Erosion
Class: Category 2
Description: Erosion of the coating from Z42 to Z44

Defect: 12

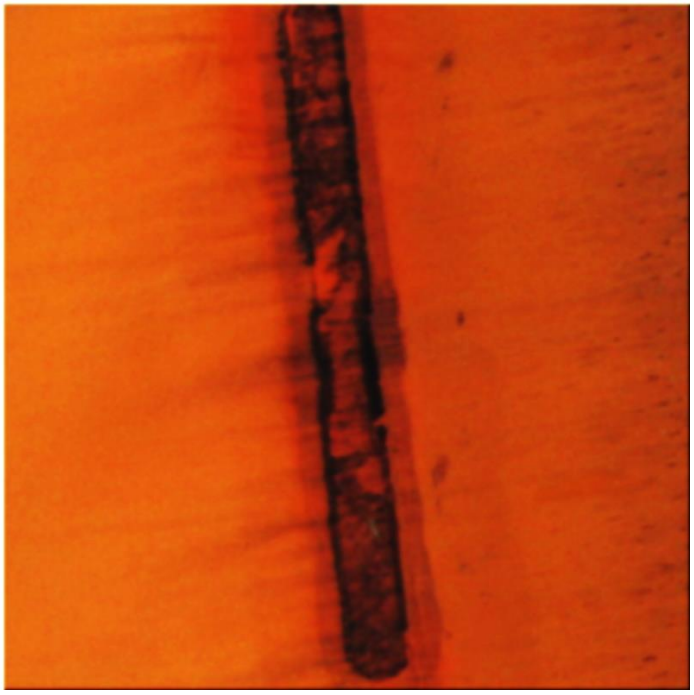


Image: 13
Blade: A
Blade Side: SS
Distance from LE: 30%
Distance from root: 40.9m
Size: 25.1cm
Type: VG missing
Class: Category 2
Description: S-size

Defect: 13

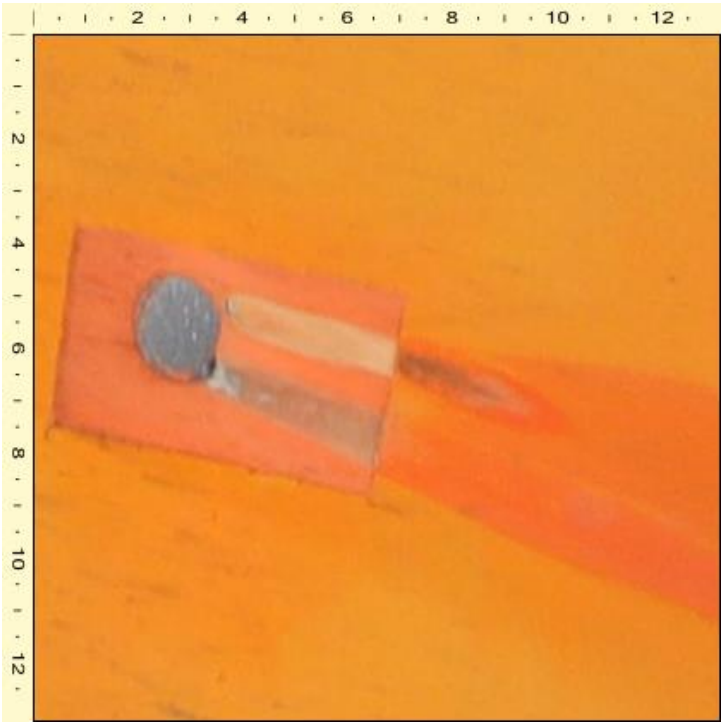
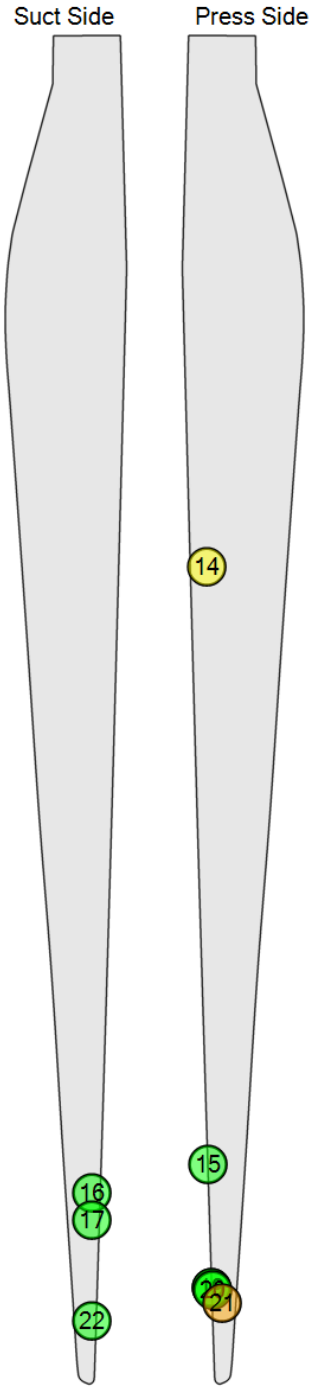


Image: 14
Blade: A
Blade Side: PS
Distance from LE: 50%
Distance from root: 44m
Size: 6.2cm
Type: Lightning discoloration
Class: Category 2
Description:

3.2 Blade B No. S2

3.2.1 Overview Blade B



Defect Number	Class	Type
14	Category 3	Lightning damage
15	Category 2	Chip in paint/coat
16	Category 2	VG missing
17	Category 2	VG damaged
18	Category 2	Chip in paint/coat
19	Category 2	Chip in paint/coat
20	Category 2	Chip in paint/coat
21	Category 4	Lightning damage
22	Category 2	VG damaged

3.2.2 Details Blade B

Defect: 14

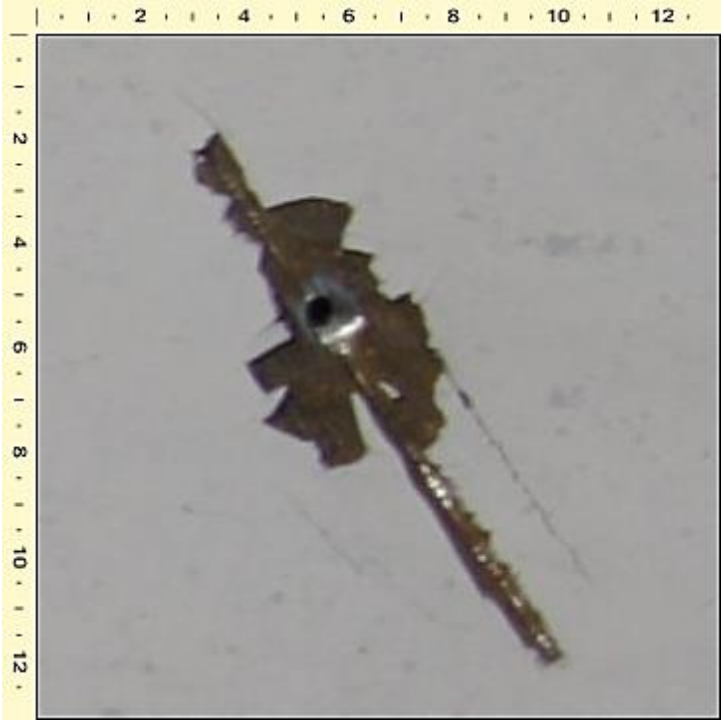


Image: 15

Blade: B

Blade Side: PS

Distance from LE: 20%

Distance from root: 17.3m

Size: 10.7 cm

Type: Lightning damage

Class: Category 3

Description: small hole within the blade shell, laminate layers are damaged

Defect: 15

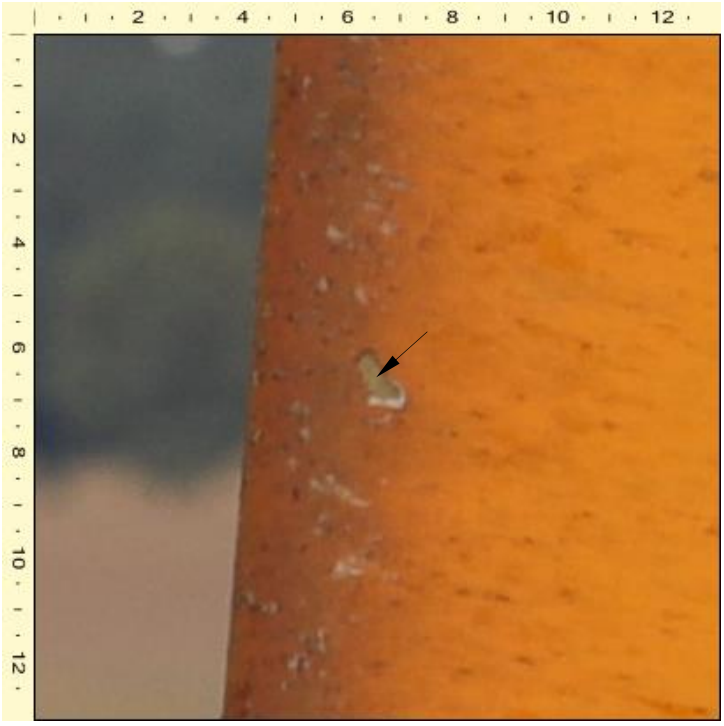


Image: 16

Blade: B

Blade Side: PS

Position: LE

Distance from root: 36.8m

Size: 1.3 cm

Type: Chip in paint/coat

Class: Category 2

Description: laminate is exposed

Defect: 16

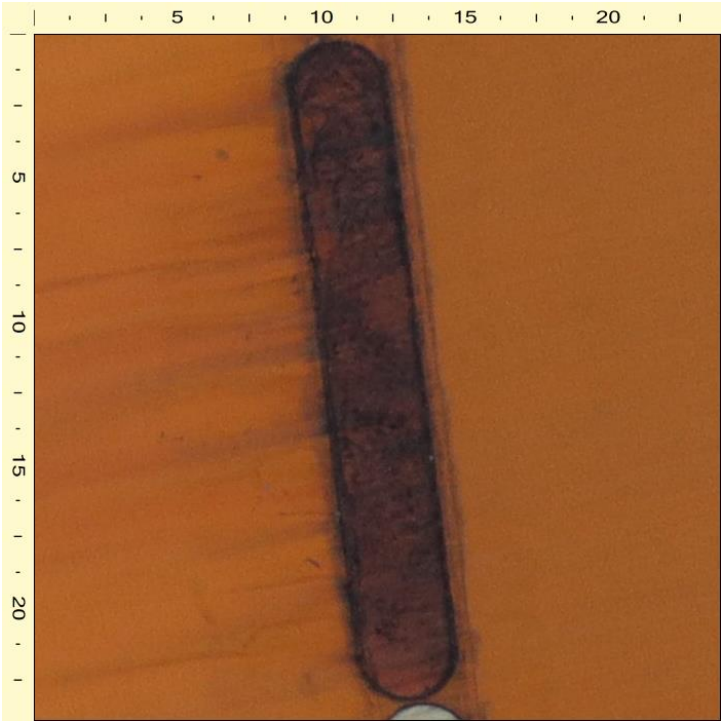


Image: 17
Blade: B
Blade Side: SS
Distance from LE: 30%
Distance from root: 37.7m
Size: 22.8 cm
Type: VG missing
Class: Category 2
Description: L-size

Defect: 17

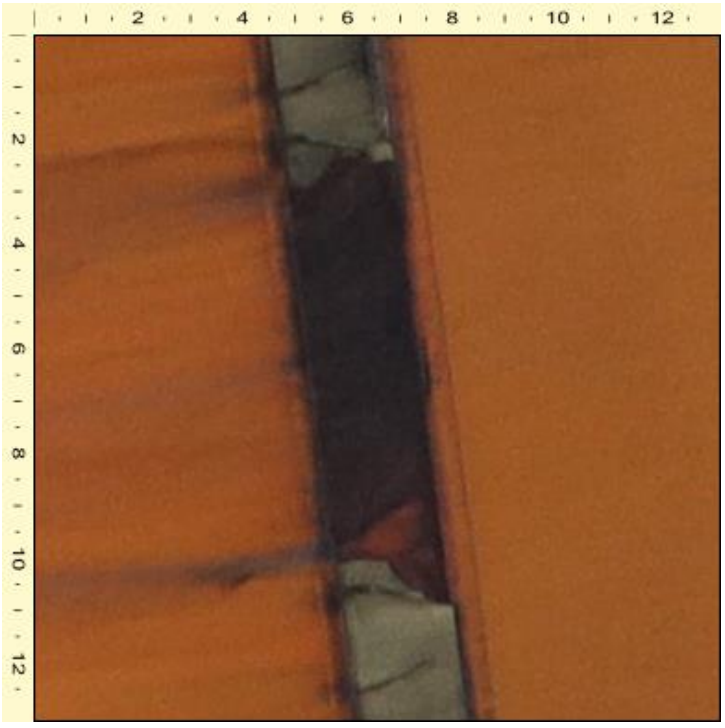


Image: 18
Blade: B
Blade Side: SS
Distance from LE: 30%
Distance from root: 38.6m
Size: 9.4 cm
Type: VG damaged
Class: Category 2
Description: S-size

Defect: 18

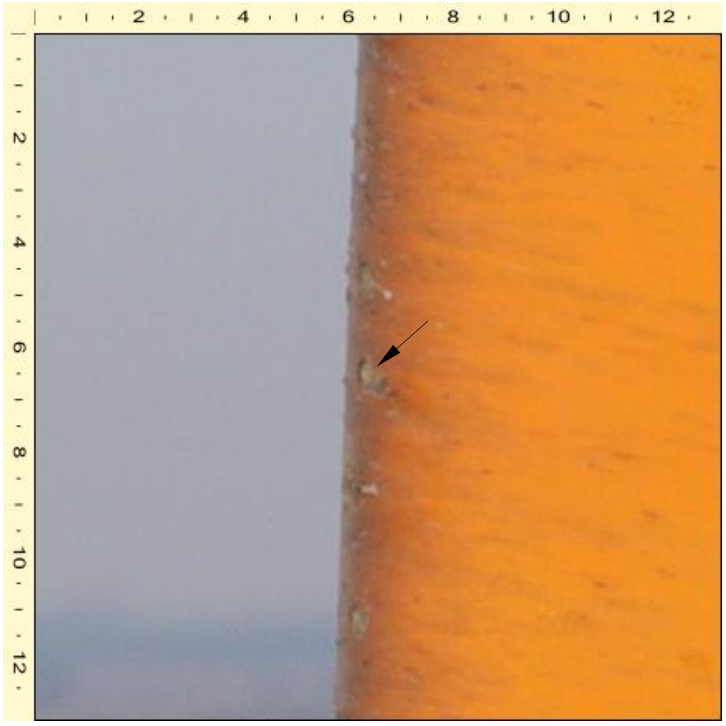


Image: 19
Blade: B
Blade Side: PS
Position: LE
Distance from root: 40.8m
Size: 0.9 cm
Type: Chip in paint/coat
Class: Category 2
Description:

Defect: 19

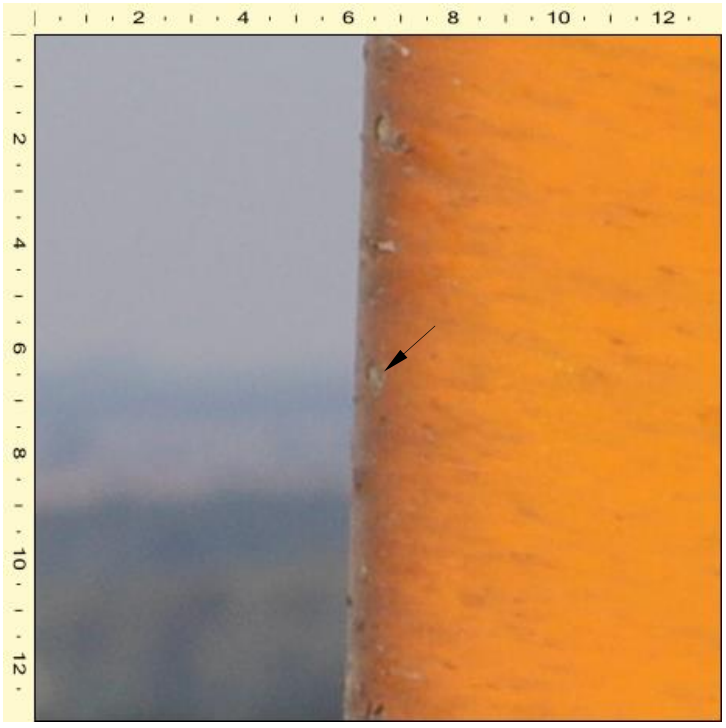


Image: 20
Blade: B
Blade Side: PS
Position: LE
Distance from root: 40.8m
Size: 0.7 cm
Type: Chip in paint/coat
Class: Category 2
Description:

Defect: 20

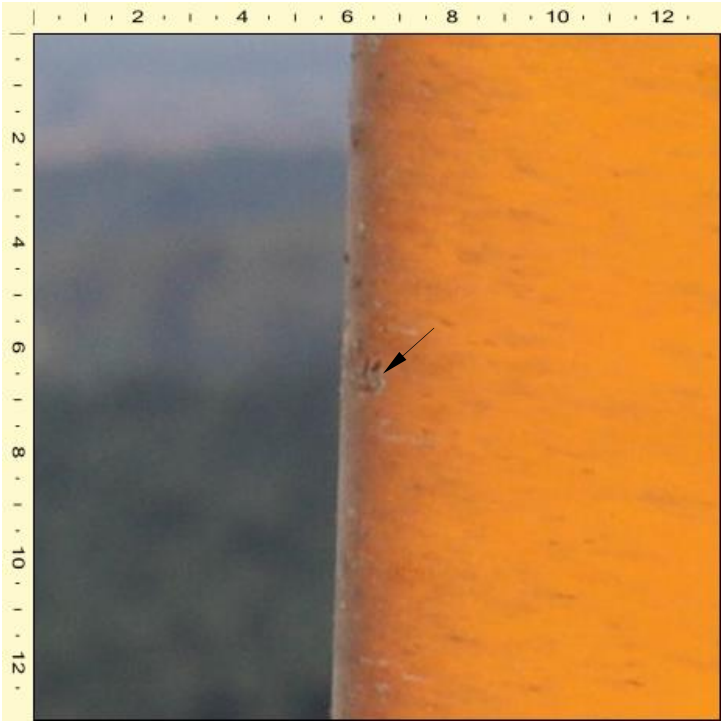


Image: 21

Blade: B

Blade Side: PS

Position: LE

Distance from root: 40.9m

Size: 0.7 cm

Type: Chip in paint/coat

Class: Category 2

Description:

Defect: 21

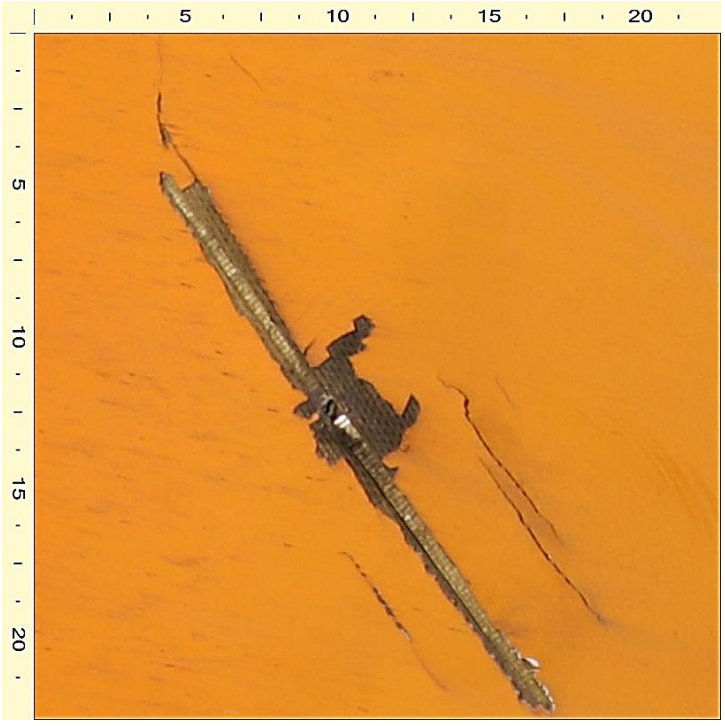


Image: 22

Blade: B

Blade Side: PS

Distance from LE: 40%

Distance from root: 43m

Size: 21.9 cm

Type: Lightning damage

Class: Category 4

Description: Damaged and cracked laminate layers

Defect: 22

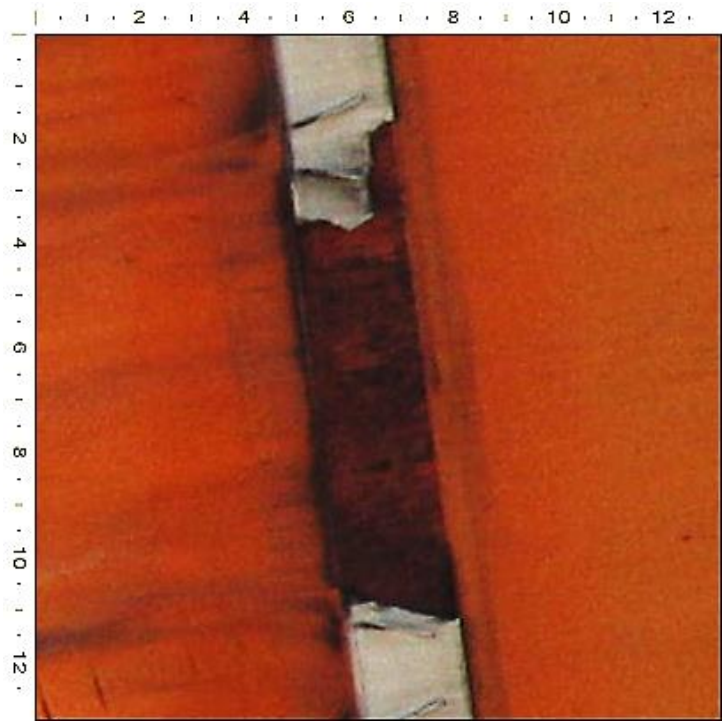
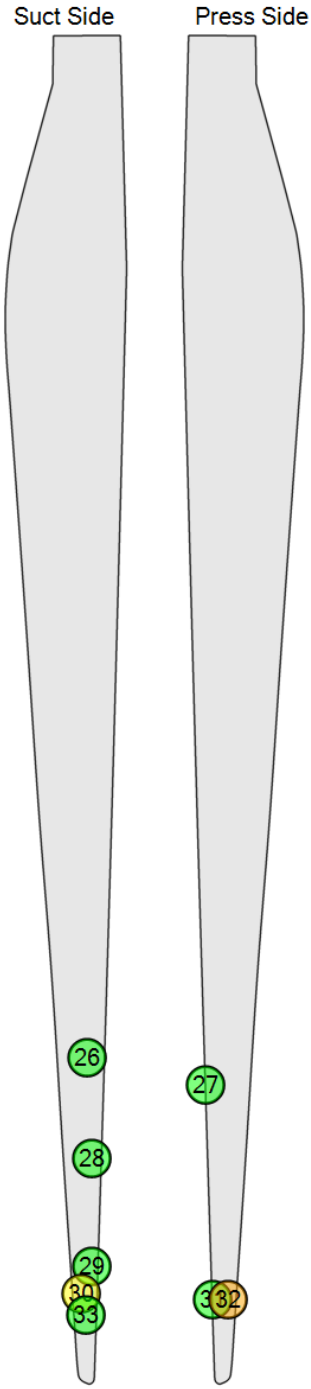


Image: 23
Blade: B
Blade Side: SS
Distance from LE: 30%
Distance from root: 41.9m
Size: 10.2 cm
Type: VG damaged
Class: Category 2
Description: S-size

3.3 Blade C No. S3

3.3.1 Overview Blade C



Defect Number	Class	Type
26	Category 2	Coat damage surface
27	Category 2	Coat damage surface
28	Category 2	VG damaged
29	Category 2	VG damaged
30	Category 3	Cracks longitudinal
31	Category 2	Coat damage surface
32	Category 4	Lightning damage
33	Category 2	Lightning discoloration

3.3.2 Details Blade C

Defect: 26

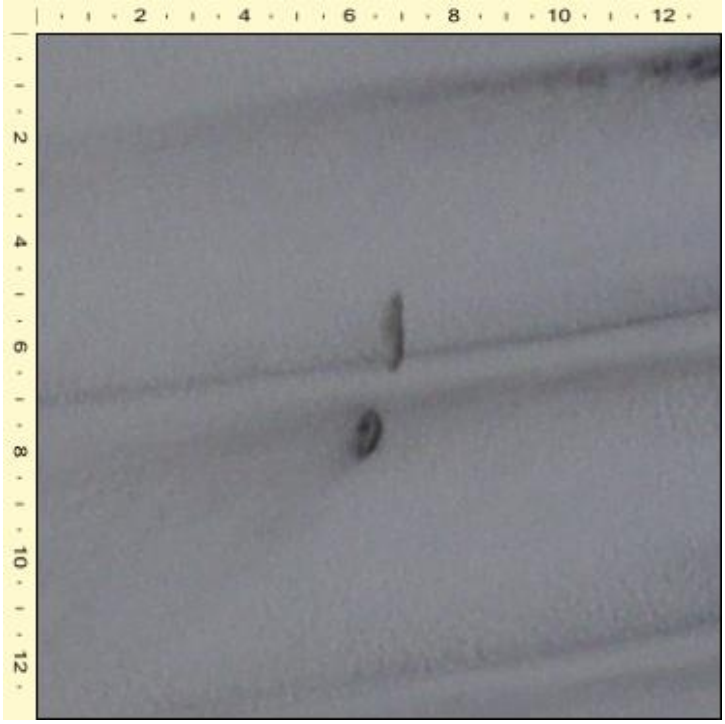


Image: 28
Blade: C
Blade Side: SS
Distance from LE: 40%
Distance from root: 33.3m
Size: 3.2 cm
Type: Coat damage surface
Class: Category 2
Description:

Defect: 27

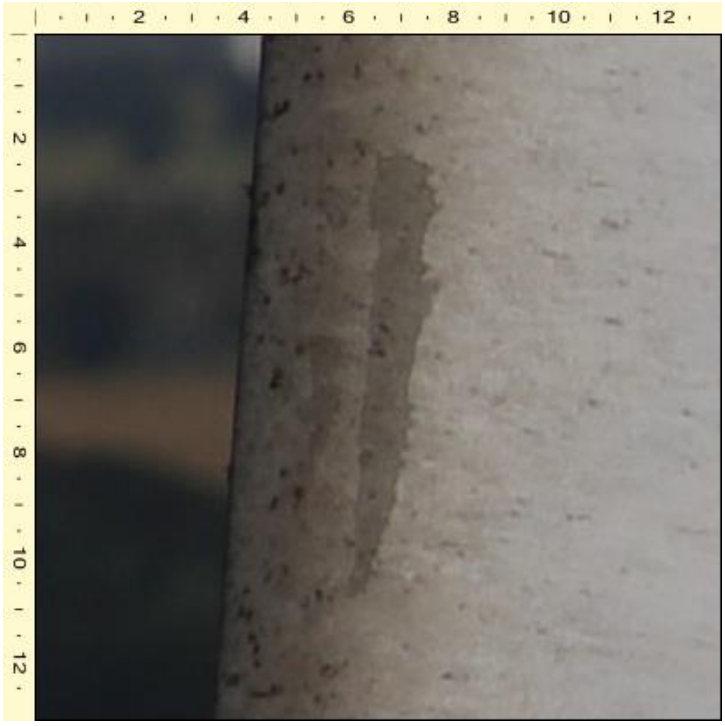


Image: 29
Blade: C
Blade Side: PS
Position: LE
Distance from root: 34.2m
Size: 9 cm
Type: Coat damage surface
Class: Category 2
Description:

Defect: 28

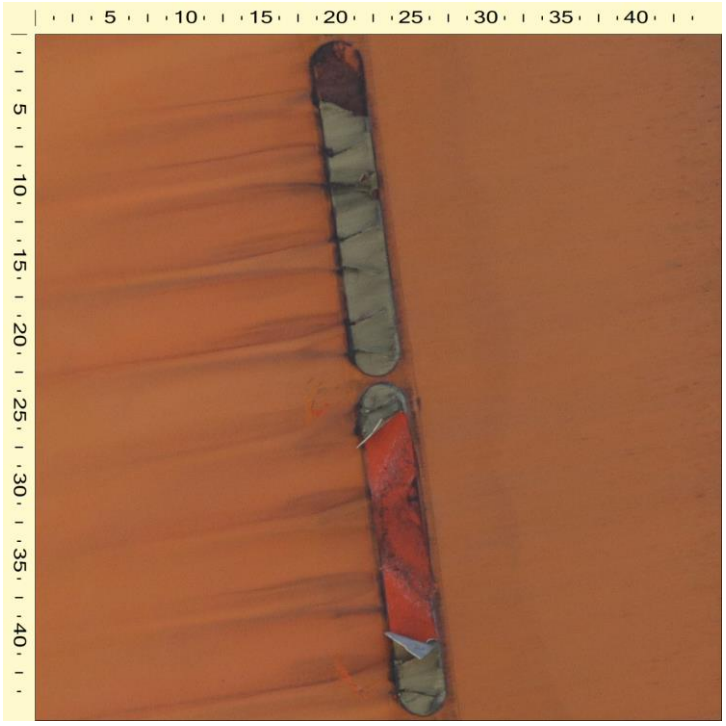


Image: 30

Blade: C

Blade Side: SS

Distance from LE: 30%

Distance from root: 36.6m

Size: 44.9 cm

Type: VG damaged

Class: Category 2

Description: L-size,
2 elements damaged

Defect: 29

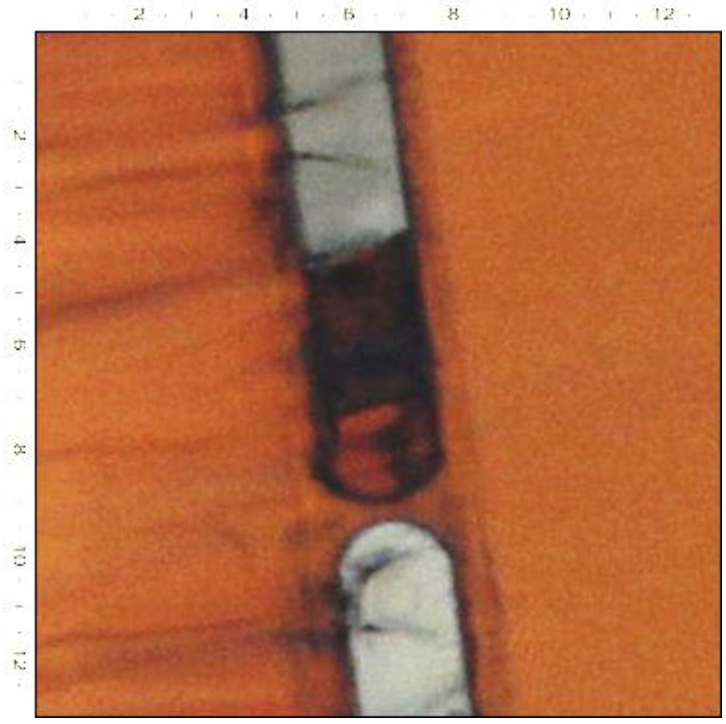


Image: 31

Blade: C

Blade Side: SS

Distance from LE: 30%

Distance from root: 40.1m

Size: 5.6 cm

Type: VG damaged

Class: Category 2

Description:

Defect: 30

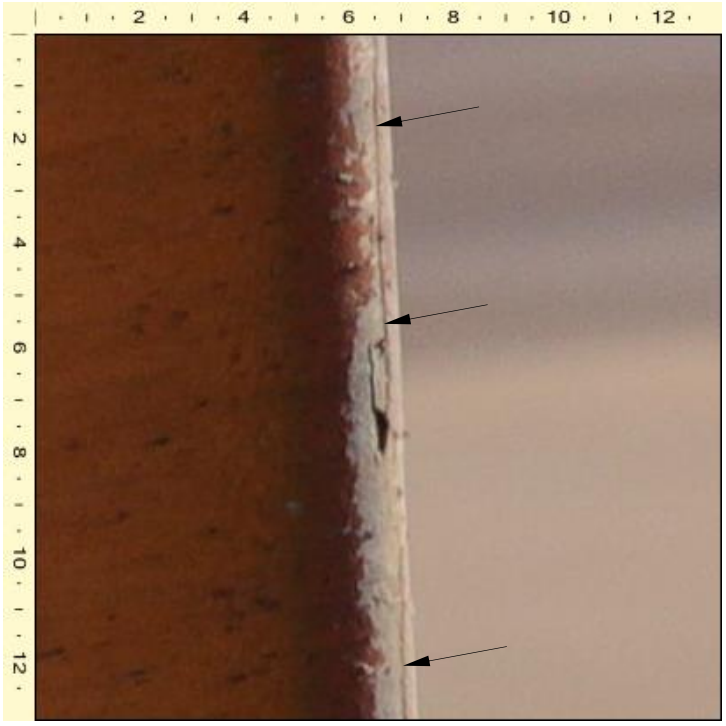


Image: 32

Blade: C

Blade Side: SS

Distance from LE: 70%

Distance from root: 43m

Size: 35.7 cm

Type: Cracks longitudinal

Class: Category 3

Description: looks like a longitudinal crack along leading-edge bonding; chip in coating

Defect: 30



Image: 33

Blade: C

Blade Side: SS

Distance from LE: 70%

Distance from root: 43m

Size: 35.7 cm

Type: Cracks longitudinal

Class: Category 3

Description looks like a longitudinal crack along leading-edge bonding; chip in coating

Defect: 31



Image: 34

Blade: C

Blade Side: PS

Position: LE

Distance from root: 43.5m

Size: 78.7 cm

Type: Coat damage surface

Class: Category 2

Description: some large coating defects from Z40 to Z44

Defect: 31



Image: 35

Blade: C

Blade Side: PS

Position: LE

Distance from root: 43m

Size: 78.7 cm

Type: Coat damage surface

Class: Category 2

Description: some large coating defects from Z40 to Z44

Defect: 31

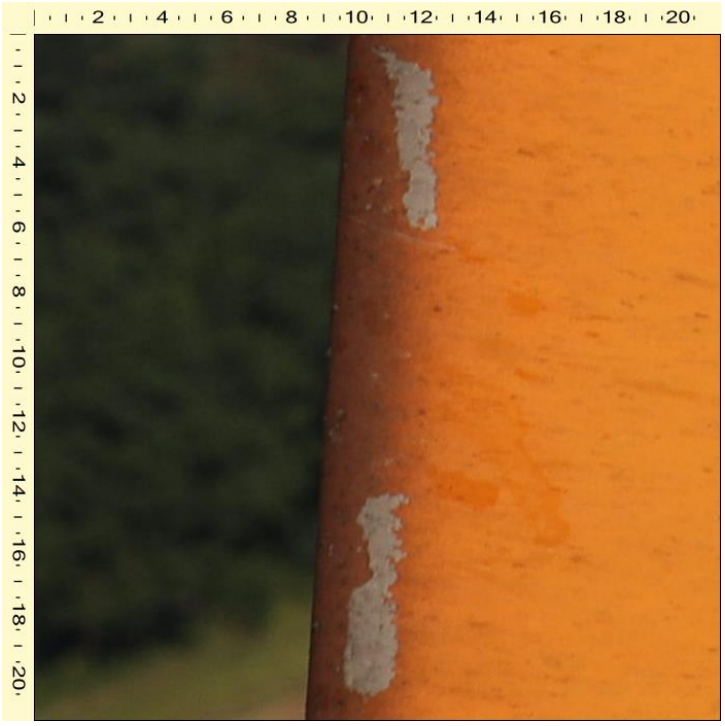


Image: 36

Blade: C

Blade Side: PS

Position: LE

Distance from root: 43m

Size: 78.7 cm

Type: Coat damage surface

Class: Category 2

Description: some large coating defects from Z40 to Z44

Defect: 32



Image: 37

Blade: C

Blade Side: PS

Distance from LE: 60%

Distance from root: 43.5m

Size: 55.5 cm

Type: Lightning damage

Class: Category 4

Description: shell laminate and trailing edge bonding are damaged

Defect: 32



Image: 38

Blade: C

Blade Side: PS

Distance from LE: 60%

Distance from root: 43.5m

Size: 55.5 cm

Type: Lightning damage

Class: Category 4

Description: shell laminate and trailing edge bonding are damaged

Defect: 33

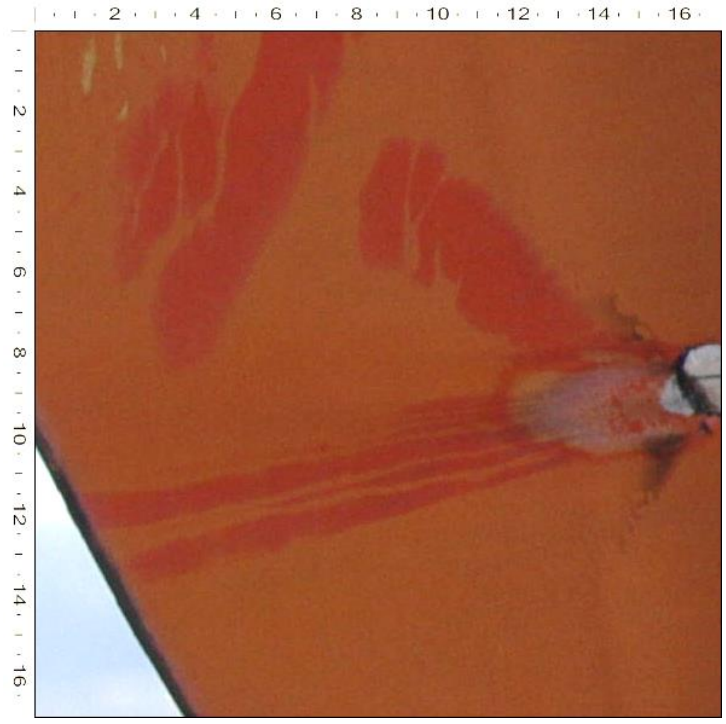


Image: 39

Blade: C

Blade Side: SS

Distance from LE: 50%

Distance from root: 44m

Size: 16.3 cm

Type: Lightning discoloration

Class: Category 2

Description: area around lightning receptor

4 Conclusions

Serious damage found on blades: A, B, C.

Blade A No. S1: Category 2

Blade B No. S2: Category 4

Blade C No. S3: Category 4

4.1 Recommendations

Repair must be done within next 3 to 6 months.

Blade A No. S1: Repair only if other damages are to be repaired.

Blade B No. S2: Repair must be done within next 3 to 6 months.

Blade C No. S3: Repair must be done within next 3 to 6 months.